

Javid Farazin

Master of physics

Education

- 2020.11 – 2024.02 Master of Physics
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran.
Supervisors: Dr. Ehsan Ahadi-Akhlaghi, Dr. Faegheh Hajizadeh
Thesis Title: **Targeted guidance of trapped particles**
- 2014.09 - 2020.07 BSc in Physics
University of Mohaghegh Ardabili (UMA), Ardabil, Iran.

Experiences

- 2024.03 – 2026.09 The Science and Technology Park
Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan, Iran.
Event Coordinator (part-time) – water and environment events
- 2022.04 – 2024.03 Aqua Solve Zanjan
Science and Technology Park, IASBS, Zanjan, Iran.
CEO: Prof. Dr. Yousef Sobouti
R&D Specialist (part-time, alongside MSc) – water desalination

Research Interests

- Optical tweezers for trapping and moving micro- and nanoparticles
- Machine learning for automating optical microscopy experiments
- Real-time closed-loop imaging and object detection in microscopy environments
- Laboratory instrumentation for biomedical imaging
- Integration of machine learning pipelines with experimental photonics hardware

Honors and Awards

- 2024.05 Best Poster Presentation Award (**First place**)
4th International Conference on Light and Light Based Technologies (ICLLT)
Gazi University, Ankara.
Title: Object Detection with Deep Learning in Optical Tweezers Applications.

Research and Executive Experience

- 2017.09 - 2020.12 Research Assistant
Faculty of Advanced Technologies, UMA, Namin, Iran.

Supervisor: Prof. Yashar Azizian

Research Areas: Fabrication of Schottky Diode Structures, Synthesis of nano-materials, Wettability measurement, Study of photocatalytic activity for photodegradation of Methylene blue from water.

Teaching Experience

- 2020.09 – 2021.01 *Faculty of Advanced Technologies, UMA, Namin, Iran.*
Modern Physics- TA
 - Teaching Assistant for problem-solving sessions
- 2019.02 – 2019.06 *Faculty of Advanced Technologies, UMA, Namin, Iran.*
Physics of light lab II- TA
 - Designed course curriculum, organized lab lectures
- 2018.09 – 2019.01 *Faculty of Advanced Technologies, UMA, Namin, Iran.*
Physics of light lab I - TA
 - Designed course curriculum, organized lab lectures

Journal Publications

- **J. Farazin**, E.A. Akhlaghi, F. Hajizadeh — Manuscript on automated optical tweezers, in preparation (2026).
- B. Akin, **J. Farazin**, Ş. Altındal, Y. Azizian-Kalandaragh “A comparison electric-dielectric features of Al/p-Si (MS) and Al/ (Al₂O₃: PVP)/p-Si (MPS) structures using voltage–current (V–I) and frequency–impedance (f–Z) measurements”; *Journal of Materials Science: Materials in Electronics* (2022).
- Y. Badali, **J. Farazin**, G. Pirgholi-Givi, Ş. Altındal, Y. Azizian-Kalandaragh “Graphene doped (Bi₂Te₃–Bi₂O₃–TeO₂): PVP dielectrics in metal–semiconductor structures”; *Applied Physics A* (2021).
- **J. Farazin**, M. Shahedi Asl, G. Pirgholi-Givi, S.A. Delbari, A. Sabahi Namini, Ş. Altındal, Y. Azizian-Kalandaragh “Effect of (Co–TeO₂-doped polyvinylpyrrolidone) organic interlayer on the electrophysical characteristics of Al/p-Si (MS) structures”; *Journal of Materials Science: Materials in Electronics* (2021).
- A. Siabi-Garjan, **J. Farazin**, G. Pirgholi-Givi, Y. Azizian-Kalandaragh “Formation of CdS/ZnS nanoparticles in polymer matrix by SILAR method: Experiments and exploring its optical properties with DDA calculations”; *Optik* (2021).
- G. Pirgholi-Givi, Y. Azizian-Kalandaragh, **J. Farazin** “Comparison of the photocatalytic activity of perovskite structures: bismuth, barium, and zinc titanate nanostructures for photodegradation of Methylene blue from water” *Journal of Photochemistry and Photobiology A: Chemistry* (2021).
- G. Pirgholi-Givi, Ş. Altındal, M. Shahedi Asl, A. Sabahi Namini, **J. Farazin**, Y. Azizian-Kalandaragh “The effect of cadmium impurities in the (PVP–TeO₂) interlayer in Al/p-Si (MS) Schottky barrier diodes (SBDs): Exploring its electrophysical parameters”; *Physica B: Condensed Matter* (2021).
- A. Sabahi Namini, M. Shahedi Asl, G. Pirgholi-Givi, S.A. Delbari, **J. Farazin**, Ş. Altındal, Y. Azizian-Kalandaragh “On the electrical characteristics of Al/p-Si diodes with and without (PVP: Sn-TeO₂) interlayer using current-voltage (I–V) measurements”; *Applied Physics A* (2020).
- Y. Azizian-Kalandaragh, **J. Farazin**, Ş. Altındal, M. Shahedi Asl, G. Pirgholi-Givi, S.A. Delbari, A. Sabahi Namini “Electrical and dielectric properties of Al/ (PVP: Zn-TeO₂) /p-Si heterojunction structures using current-voltage (I–V) and impedance-frequency (Z–f) measurements”; *Applied Physics A* (2020).

- S. Fakhri-Mirzanagh, K. Ahadzadeh-Namin, G. Pirgholi Givi, **J. Farazin**, Y. Azizian-Kalandaragh "The effect of capping agent on the structural, optical properties and photocatalytic activity of MgO nanostructures"; Physica B: Condensed Matter (2020).
- Ş. Altındal, **J. Farazin**, G. Pirgholi-Givi, E. Marıl, Y. Azizian-Kalandaragh "The effects of (Bi₂Te₃–Bi₂O₃–TeO₂-PVP) interfacial film on the dielectric and electrical features of Al/p-Si (MS) Schottky barrier diodes (SBDs)"; Physica B: Condensed Matter (2020).
- **J. Farazin**, G. Pirgholi-Givi, Y. Azizian-Kalandaragh "Wettability measurement, optical characteristics, and investigation of the quantum confinement effect of ZnS-scotch tape nanocomposite films prepared by successive ionic layer adsorption and reaction (SILAR) method"; Physica B: Condensed Matter (2019).

Conference Papers

- **J. Farazin**, M. Rahmani, F. Hajizadeh, E. Ahadi Akhlaghi, "Object Detection with Deep Learning in Optical Tweezers Applications"; in the 4th International Conference on Light and Light-based Technologies (ICLLT-2024), Ankara, Turkey, **2024**.
- Y. Azizian-Kalandaragh, **J. Farazin**, G. Pirgholi-Givi "Preparation of CdS/ZnS-polymer nanocomposite by SILAR method"; in the twenty-sixth Iranian Conference on Optics and Photonics (ICOP 2020), Karaj, Iran, 2020.
- B. Akin, G. Pirgholi-Givi, **J. Farazin**, Y. Azizian-Kalandaragh, Ş. Altındal "Utilization of Al₂O₃/PVP nanocomposite as an interfacial layer for Schottky structures"; in 2019 Thirteenth International Congress on Artificial Materials for Novel Wave Phenomena (Metamaterials), pp. X-017. IEEE, 2019.
- Ş. Altındal, **J. Farazin**, G. Pirgholi-Givi, Y. Azizian-Kalandaragh, B. Akin "Morphological and Structural Properties of Metamaterial Based on ITO/Sapphire/ZnS/Al Superlattice"; in 2019 International Conference on Electromagnetics in Advanced Applications (ICEAA), Granada, Spain, pp. 0816-0819. IEEE, 2019.
- Y. Azizian-Kalandaragh, G. Pirgholi-Givi, **J. Farazin** "The Effect of the Etching Rate on the Main Electrical Properties of p-Si"; In International Natural Science, Engineering and Material Technologies Conference (NEM), Istanbul, Turkey, pp. 156. Abstract Book, 2019.
- G. Pirgholi-Givi, **J. Farazin**, Y. Azizian-Kalandaragh, L. Anatolievich-Golovan "The Effect of the Electrochemical Etching Prosses Rate on Electrical Properties of P-Silicon"; In first International Conference on Optoelectronics, Applied Optics and Microelectronics, Namin, Iran, pp. 113-117. Full text Book, 2019.
- Y. Azizian-Kalandaragh, **J. Farazin**, G. Pirgholi-Givi; "Optical and structural characteristics of ZnS-polymer nanocomposite materials prepared by SILAR method"; In Fifth International Conference on Materials Science and Nanotechnology for Next Generation (MSNG), Cappadocia, Turkey, pp. 5. Abstract Book, 2018.
- Y. Azizian-Kalandaragh, **J. Farazin**, G. Pirgholi-Givi, "Preparation and Optical Characterization of CdS/ZnS polymer Superlattice Nanocomposites"; in Third International Conference on Organic Electronic Material Technologies (OEMT), Kirklareli, Turkey, pp. 120. Abstract Book, 2018.

Skills

Computer Skills

- Programming:
 - Python (Expert)
 - Python Libraries: Numpy, Scipy, CV2, PIL, Pandas, scikit-learn, TensorFlow (Expert)
 - MATLAB (Experienced)
- Data science and specialization:
 - Machine Learning (Experienced)
 - Deep Learning (Experienced)
 - Reinforcement Learning (Experienced)

Lab Skills

- Optical tweezers — full system assembly, alignment, calibration and operation (Expert)
- Bright-field and fluorescence microscopy; real-time image acquisition and analysis (Expert)
- Nanostructure synthesis and characterization; FTIR and optical spectroscopy (Proficient)
- Spectroscopy (Proficient)

Languages

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| ❖ English – C1 | ❖ Turkish - Native |
| ❖ Azerbaijani - Mother Tongue | ❖ Persian - Native |